

17. SUM OF INTEGERS

AIM

To write a Prolog program to find the sum of integers from 1 to N using recursion.

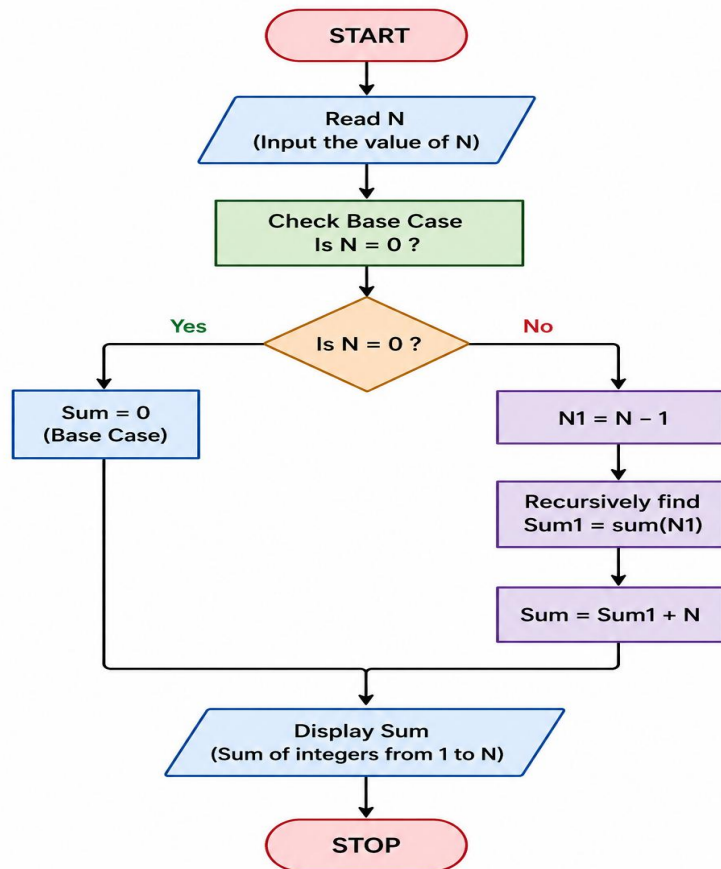
OBJECTIVE

- To understand recursion in Prolog.
- To calculate the sum of integers from 1 to N.
- To implement the program using Prolog predicates.

ALGORITHM

1. Start.
2. Read the value of N.
3. Set the base case: $\text{sum}(0,0)$.
4. For $N > 0$, calculate $N1 = N - 1$.
5. Recursively calculate the sum up to N1.
6. Add N to the previous sum.
7. Display the final sum.
8. Stop.

Flowchart:



CODE:

sum(0, 0).

sum(N, Sum) :-

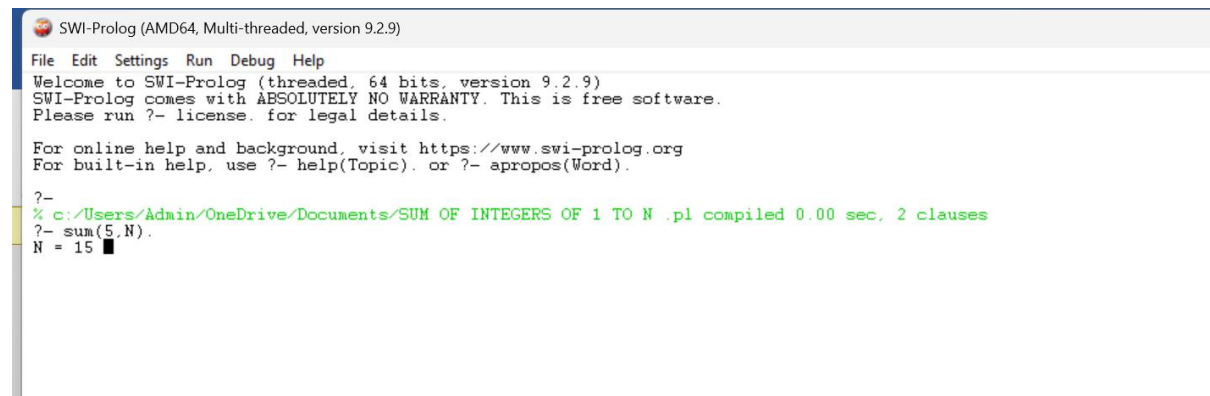
N > 0,

N1 is N - 1,

sum(N1, Sum1),

Sum is Sum1 + N.

OUTPUT:



```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/Admin/OneDrive/Documents/SUM OF INTEGERS OF 1 TO N .pl compiled 0.00 sec, 2 clauses
?- sum(5,N).
N = 15
```

RESULT:

Thus, the Prolog program to calculate the sum of integers from 1 to N was successfully implemented and executed using recursion.